Simple Pendulum - Example 5.17

Reference: Applied Mechanics: Reed's Vol. 2

Ex. 5.17: Find the length of a simple pendulum to make:

1. one complete oscillation per second
2. half an oscillation per second

The period of a simple pendulum is:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Rearranging for length L:

$$\boxed{L = \frac{gT^2}{4\pi^2}}$$

Since frequency $f = 1/T$, this can also be written as:

$$L = \frac{g}{4\pi^2 f^2}$$

Part (i) — One Complete Oscillation per Second

$$f = 1\ \text{Hz} \implies T = 1\ \text{s}$$

$$L = \frac{9.81 \times 1^2}{4\pi^2} = \frac{9.81}{39.478}$$

$$\boxed{L = 0.2485\ \text{m} \approx 248.5\ \text{mm}}$$

Part (ii) — Half an Oscillation per Second

$$f = 0.5 \text{ Hz} \implies T = 2 \text{ s}$$

$$L = \frac{9.81 \times 2^2}{4\pi^2} = \frac{9.81 \times 4}{39.478}$$

$$\boxed{L = 0.9940 \text{ m} \approx 994 \text{ mm}}$$

Final Answers

Doubling the period requires **four times** the length, since $L \propto T^2$.

Case	Frequency f	Period T	Length L
(i)	1 Hz	1 s	248.5 mm
(ii)	0.5 Hz	2 s	994.0 mm

Python Solution

```
import math

g = 9.81  # m/s^2

for f in [1.0, 0.5]:
    T = 1 / f
    L = (g * T**2) / (4 * math.pi**2)
    print(f"f = {f} Hz  →  T = {T:.1f} s  →  L = {L*1000:.2f} mm")
```